

3DCrewCap: Applying 3D Volumetric Video Capture for XR Helicopter Rescue Crew Training and Simulation

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ABSTRACT

Helicopter rescue crew training scenarios are complex and hard to simulate in game engines with animated 3D digital characters. This challenge is compounded when simulating Realtime photorealistic animated character sequences on XR-based Head-Mounted Displays (HMD). In this research, we present a practical early-stage development of 3D volumetric video capture and playback workflow for use in helicopter rescue crew training on XR HMDs. We break down the workflow of using Gaussian Splat approaches to construct keyframed 3D animated models of the rescue crew training actions. This novel application approach provides a practical example of the photorealistic capture and XR display of helicopter rescue crew performing training scenarios.

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1 INTRODUCTION

Extended reality (XR) simulations have been increasingly utilised in helicopter rescue crew training as a safer and more cost-effective alternative to live training whilst being less restricted by time and space [Sun et al., 2020]. A key factor in XR simulation training is the level of fidelity, that is, the extent to which the simulation is able to recreate the real-life scenario of training [Stone, 2012]. Existing literature has widely acknowledged the importance of maintaining a sufficient level of fidelity in the simulation to ensure effective training [Stevens et al., 2015].

However, virtual reconstructed 3D simulations with high levels of visual and behavioural fidelity such as game engine characters are usually demanding in time, resource and cost to create, especially when simulating subtle and non-verbal cues [Hackett et al., 2022]. 3D volumetric video signals a promising solution by allowing the creator to capture the actor and actions with multiple cameras



Figure 1: – Example Output a) Capture of the ‘ready for recovery’ crew action b) Final Output XR 3D volumetric video screenshot

from multiple angles simultaneously and integrate the synthesised volumetric footage into the virtual scenes. This approach enables an efficient and accurate recreation of the real-world training session, whilst allowing the trainee to engage with the captured content in an immersive context [O’Dwyer et al., 2020].

Conventional methods used in 3D volumetric video such as meshes and point clouds can be problematic with considerable data size and limited accuracy [Lin et al., 2023]. However, with the advent of Radiance Field techniques such as Gaussian Splats [Kerbl et al., 2023], the capture is processed holistically with its size optimised whereas high resolution retained. In this research we explore workflows that utilise Gaussian Splats in 3D volumetric video capture and playback for helicopter rescue crew training.

2 DESIGN AND IMPLEMENTATION

In order to capture authentic training scenarios, seven discreet crew actions and movements were identified for 3D volumetric capture. These were i) ready for recovery; ii) hooked on, ready to winch; iii) winch down; iv) winch up; v) hold at this height; vi) stop all movement; and vii) disconnect. Figure 1 shows an example of action i) ready for recovery.

2.1 Performance Capture

The first stage of our workflow is to capture the crew member’s actions. This uses the 3DXLab capture arena consisting of 26 IO Industries 89B93CV cameras each recording at a resolution of 2176x4096 pixels. The cameras are arranged in 8 columns of 3, with 2 extra cameras recording from above. Using 12mm lenses gives a working

capture volume of approximately 2m diameter. All cameras have their shutters synced $\rightarrow$ via PTP to make sure they are recording frames at the same time. This performance was recorded at 60fps, and subsequently 30fps was selected for playback as a compromise between playback media size, gaussian splat processing time and video quality. The capture arena was calibrated by using 23 calibration markers placed on the trusses and floor of the arena. The images from the cameras containing these markers are processed by the RealityCapture 1.3 software package [Capturing Reality s.r.o., 2023] which calculates the position and orientation of each camera. This calibration data was used further on in the workflow by the gaussian splat training algorithms. In the test capture of a crew member illustrated in Figure 1 (a) the capture time was 54 seconds at 60fps. This created 286GB of compressed raw image data from the cameras, which extracted out to 918GB of PNG images.

2.2 Data Processing and Gaussian Splat Generation

The second stage of our workflow involved processing each frame that was recorded individually. This allowed for simultaneous frame processing on multiple nodes of 3DXLab render farm. See Figure 2 - Workflow Overview Diagram. The process was as follows:

- Take the 26 camera images and camera alignment information from the calibration step into RealityCapture.
- Export the undistorted images and sparse point cloud from RealityCapture into bundler format.
- Convert the bundler format into COLMAP format.
- Train the Gaussian Splat using the code based on [Kerbl et al., 2023].
- Crop the splat to remove all background, leaving only the subject (and some floor around them). The floor can be removed in the rendering using chroma key techniques.

Each frame was trained for 15,000 iterations. We used our lab GPU render farm with 16 Dell PC workstations each with 1x NVIDIA A6000 graphics card. This took approximately 28 minutes per frame, or 756 render hours for the above take (54 seconds, 1620 frames at 30fps). After cropping, each splat frame was approximately 14MB in size and contained $\sim$ 60,000 gaussians.

2.3 XR Implementation – Gaming Engine Unity3D

The final stage of our workflow is to insert the 3D animated volumetric avatars into a multi-user mixed-reality (XR) training. See Figure 1 (b). Varjo XR-3 HMDs were employed with high quality video passthrough, allowing participants to see and interact with each other while also viewing the virtual world that implements the simulated environment. Unity3D was used as the game engine to prototype this XR content. The 3D volumetric avatars can be overlayed over the real-world view as well as placed within the virtual world, depending on the need of the interaction, such as piloting the aircraft or signalling from the ground. The modular implementation of the application allows for variable number of users, with virtual avatars used in place of absent participants.

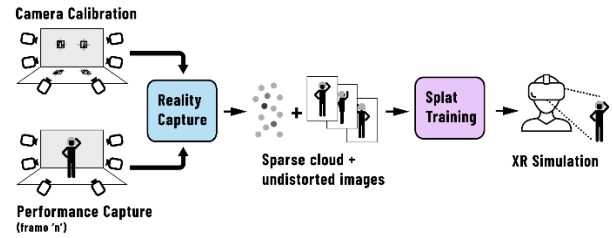


Figure 2: – Workflow Overview Diagram

3 DISCUSSION

The novelty of this early-stage research lies in the application of 3D volumetric video capture and XR display in helicopter rescue crew training scenarios. We outline a practical workflow with a 26-camera capture arena and GPU farm to construct Gaussian Splat key frame animation driven 3D photorealistic crew avatars. There are limitations to this approach, firstly the output is a Gaussian Splat frame-by-frame sequence. This technique has no optimisation across time. Secondly uncaptured animations and interactions within the simulated scenarios are currently not possible. Therefore, a novel pose capability is not possible with our workflow. We see opportunities for future work. These include the implementation of parametric 4D approaches to improve the Gaussian Splat representation across 4D performances [Lin et al., 2023], and the application of novel-pose synthesis and novel-view synthesis [Kocabas et al., 2023] to allow for dynamic unrecorded interactions with the volumetric avatars.

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